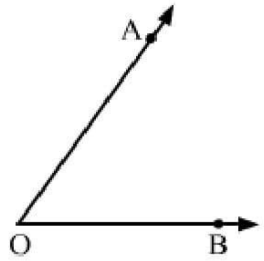
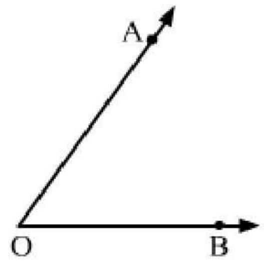
Exercise 7A

Solution 1:

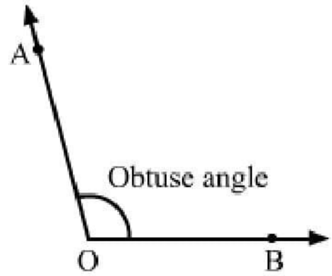
(i) Angle – When two rays originate from the same end point, then an angle is formed.



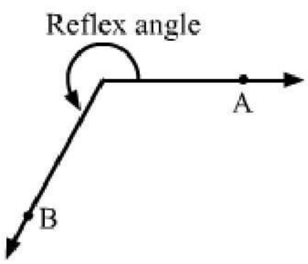
(ii) Interior of an angle – The interior of an angle is the set of all points in its plane, which lie on the same side of OA as B and also on the same side of OB as A.



(iii)Obtuse angle – An angle whose measure is more than 90° but less than 180° is called an obtuse angle.



(iv) Reflex angle – An angle whose measure is more than 180° but less than 360° is called a reflex angle.



(v) Complementary angles – Two angles are said to be complementary, if the sum of their measure is 90°.

(vi) Supplementary angles – Two angles are said to be supplementary if the sum of their measures is 180°.

Solution 2:

(i) Complement of $55° = 90° - 55° = 35°$

(ii) Complement of $16° = 90° - 16° = 74°$

(iii) complement of $90° = 90° - 90° = 0°$

(iv) $\frac{2}{3}$ of a right angle $= \frac{2}{3} \times 90° = 60°$

Now, complement of $\frac{2}{3}$ right angle $= 90° - 60° = 30°$

Solution 3:

(i) Supplement of $42° = 180° - 42° = 138°$

(ii) Supplement of $90° = 180° - 90° = 90°$

(iii) Supplement of $124° = 180° - 124° = 56°$

(iv) $\frac{3}{5}$ of a right angle $= \frac{3}{5} \times 90° = 54°$

Now, supplement of $\frac{3}{5}$ right angle $= 180° - 54° = 126°$

Solution 4:

(i) Let the measure of the required angle be x°.

Then, in case of complementary angles:

$x + x = 90°$

$\Rightarrow 2x = 90°$

$\Rightarrow x = 45°$

Hence, measure of the angle that is equal to its complement is 45°.

(ii) Let the measure of the required angle be x°.

Then, in case of supplementary angles:

$x + x = 180°$

$\Rightarrow 2x = 180°$

$\Rightarrow x = 90°$

Hence, measure of the angle that is equal to its supplement is 90°.

Solution 5: Let the measure of the required angle be x°.

Then, measure of its complement = $(90 - x)°$

Therefore,

$x - (90° - x) = 36°$

$\Rightarrow x - 90° + x = 36°$

$\Rightarrow 2x = 36° + 90°$

$\Rightarrow 2x = 126°$

$\Rightarrow x = 63°$

Hence, the measure of the required angle is 63°

Solution 6: Let the measure of the angle be x°.

$\therefore$ Supplement of $x° = 180° - x°$

It is given that,

$(180° - x°) - x° = 30°$

$\Rightarrow 180° - 2x° = 30°$

$\Rightarrow 2x° = 180° - 30°$

$\Rightarrow 2x° = 150°$

$\Rightarrow x° = 75°$

Thus, the measure of the angle is 75°.

Solution 7: Let the measure of the required angle be x.

Then, measure of its complement $=(90°-x)4$

Therefore,

$x = (90° - x)\ 4$

$\Rightarrow x = 360° - 4x$

$\Rightarrow 5x = 360°$

$\Rightarrow x = 72°$

Hence, the measure of the required angle is 72°.

Solution 8: Let the measure of the required angle be x.

Then, measure of its supplement $= (180° - x)$

Therefore,

$x = (180° - x)5$

$\Rightarrow x = 900° - 5x$

$\Rightarrow 6x = 900°$

$\Rightarrow x = 150°$

Hence, the measure of the required angle is 150°.

Solution 9: Let the measure of the required angle be x°.

Then, measure of its complement $=(90-x)°$

And, measure of its supplement$=(180-x)°$

Therefore,

$(180 - x) = 4(90 - x)$

$\Rightarrow 180 - x = 360 - 4x$

$\Rightarrow 3x = 180$

$\Rightarrow x = 60$

Hence, the measure of the required angle is 60°.

Solution 10: Let the measure of the required angle be x°.

Then, the measure of its complement $= (90-x)°$

And the measure of its supplement $= (180 - x)°$

Therefore,

$$(90 - x)° = \frac{1}{3}(180 - x)°$$

$\Rightarrow 3\,(90 - x)° = (180 - x)°$

$\Rightarrow 270° - 3x° = 180° - x°$

$\Rightarrow 2x = 90°$

$\Rightarrow x = 45°$

Hence, the measure of the required angle is 45°.

Solution 11: Let the two angles be $4x$ and $5x$, respectively.

Then,

$4x + 5x = 90°$

$\Rightarrow 9x = 90°$

$\Rightarrow x = 10°$

Hence, the two angles are $4x = 4 \times 10° = 40°$ and $5x = 5 \times 10° = 50°$

Solution 12: Two angles whose sum is 90° are called complementary angles.

It is given that the angles $(2x - 5)°$ and $(x - 10)°$ are the complementary angles.

$\therefore (2x - 5)° + (x - 10)° = 90°$

$\Rightarrow 3x° - 15° = 90°$

$\Rightarrow 3x° = 90° + 15°$

$\Rightarrow 3x° = 105°$

$\Rightarrow x° = \dfrac{105°}{3}$

$\Rightarrow x° = 35°$

Thus, the value of x is 35.

Exercise 7B

Solution 1: From the figure we know that ∠AOC and ∠BOC are a linear pair of angles

So, we get

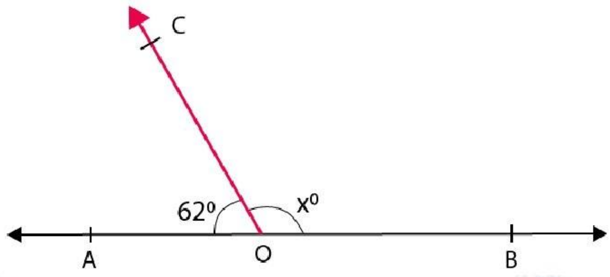
$\angle AOC + \angle BOC = 180^o$

$\Rightarrow 62^o + x^o = 180^o$

$\Rightarrow x^o = 180^o - 62^o$

$\Rightarrow x^o = 118^o$

Therefore, the value of x is 118^o.



Solution 2: As AOB is a straight line, the sum of angles on the same side of AOB, at a point O on it, is 180°.

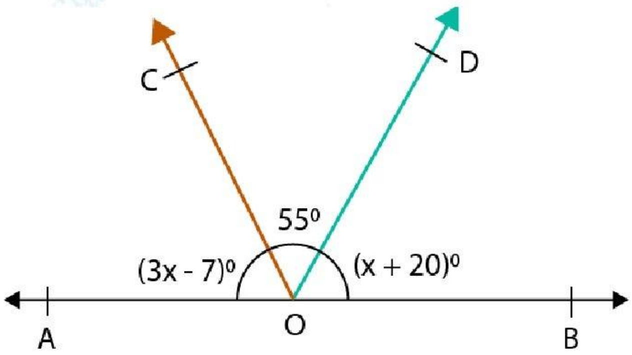
Therefore,

$\angle AOC + \angle COD + \angle BOD = 180°$

$\Rightarrow (3x - 7)° + 55° + (x + 20)° = 180°$

$\Rightarrow 4x = 112°$

$\Rightarrow x = 28°$



Hence,

$\angle AOC = 3x - 7$

$= 3 \times 28 - 7$

$= 77°$

and $\angle BOD = x + 20$

$= 28 + 20$

$= 48°$

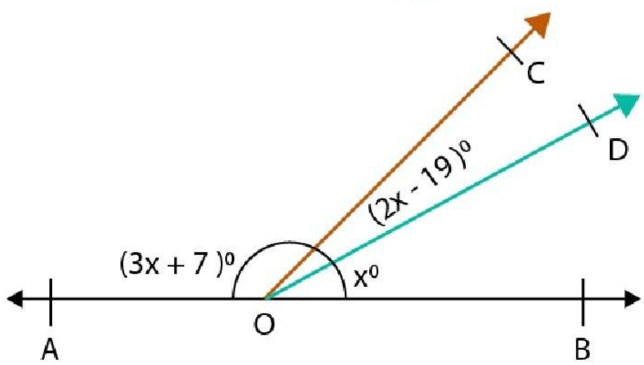
Solution 3: AOB is a straight line. Therefore,

$\angle AOC + \angle COD + \angle BOD = 180°$

$\Rightarrow (3x + 7)° + (2x - 19)° + x° = 180°$

$\Rightarrow 6x = 192°$

$\Rightarrow x = 32°$



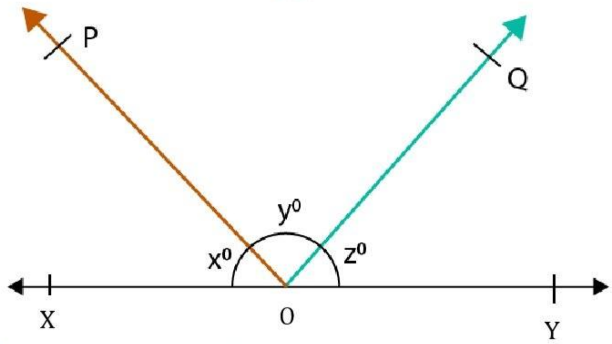
Therefore,

$\angle AOC = 3 \times 32° + 7 = 103°$

$\angle COD = 2 \times 32° - 19 = 45°$ and

$\angle BOD = 32°$

Solution 4: Let $x = 5a$, $y = 4a$ and $z = 6a$



XOY is a straight line. Therefore,

$\angle XOP + \angle POQ + \angle YOQ = 180°$

$\Rightarrow 5a + 4a + 6a = 180°$

$\Rightarrow 15a = 180°$

$\Rightarrow a = 12°$

Therefore,

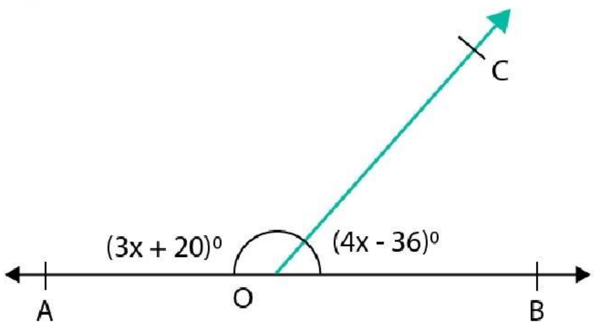
$x \Rightarrow 5 \times 12° = 60°$

$y \Rightarrow 4 \times 12° = 48°$ and

$z \Rightarrow 6 \times 12° = 72°$

Solution 5: We know that AOB will be a straight line only if the adjacent angles form a linear pair.

$\angle BOC + \angle AOC = 180°$



$\Rightarrow (4x - 36)° + (3x + 20)° = 180°$

$\Rightarrow 4x - 36° + 3x + 20° = 180°$

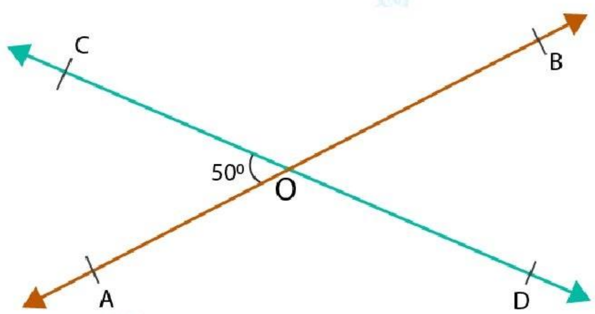
$\Rightarrow 7x = 180° - 20° + 36°$

$\Rightarrow 7x = 196°$

$\Rightarrow x = 28°$

Therefore, the value of x is $28°$.

Solution 6: We know that if two lines intersect then the vertically-opposite angles are equal.



Therefore,

$\angle AOC = \angle BOD = 50°$

Let $\angle AOD = \angle BOC = x°$

Also, we know that the sum of all angles around a point is 360°

Therefore,

$\angle AOC + \angle AOD + \angle BOD + \angle BOC = 360°$

$\Rightarrow 50 + x + 50 + x = 360°$

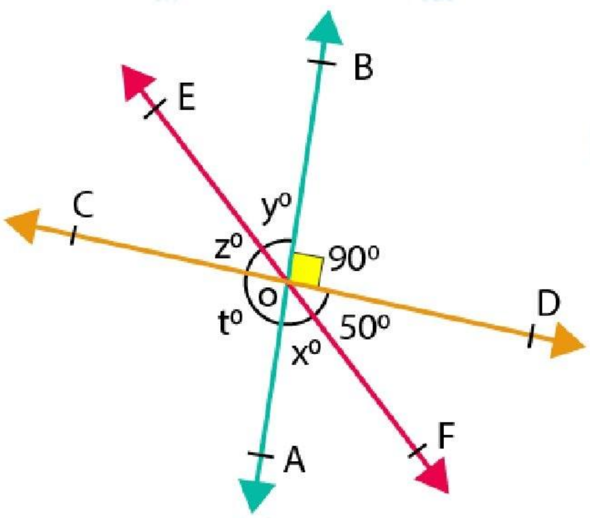
$\Rightarrow 2x = 260°$

$\Rightarrow x = 130°$

Hence, $\angle AOD = \angle BOC = 130°$

Therefore, $\angle AOD = 130°$, $\angle BOD = 50°$ and $\angle BOC = 130°$

Solution 7: We know that if two lines intersect, then the vertically opposite angles are equal.



$\therefore \angle BOD = \angle AOC = 90°$

Hence, $t = 90°$

Also,

$\angle DOF = \angle COE = 50°$

Hence, $z = 50°$

Since, AOB is a straight line, we have:

$\angle AOC + \angle COE + \angle BOE = 180°$

$\Rightarrow 90 + 50 + y = 180°$

$\Rightarrow 140 + y = 180°$

$\Rightarrow y = 40°$

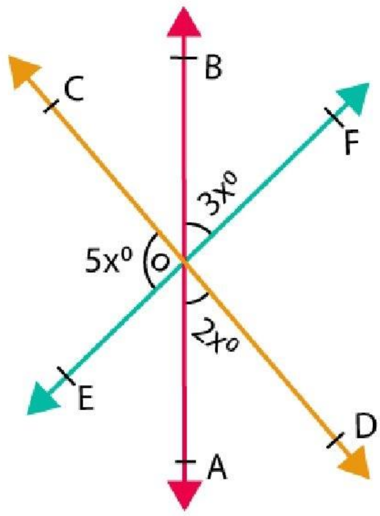
Also,

$\angle BOE = \angle AOF = 40°$

Hence, $x = 40°$

$\therefore x = 40°, y = 40°, z = 50°$ and $t = 90°$

Solution 8: We know that if two lines intersect, then the vertically-opposite angles are equal.



$\therefore \angle DOF = \angle COE = 5x°$

$\angle AOD = \angle BOC = 2x°$ and

$\angle AOE = \angle BOF = 3x°$

Since, AOB is a straight line, we have:

$\angle AOE + \angle COE + \angle BOC = 180°$

$\Rightarrow 3x + 5x + 2x = 180°$

$\Rightarrow 10x = 180°$

$\Rightarrow x = 18°$

Therefore,

$\angle AOD = 2 \times 18° = 36°$

$\angle COE = 5 \times 18° = 90°$

$\angle AOE = 3 \times 18° = 54°$

Solution 9: Let the two adjacent angles be 5x and 4x.

Then,

$5x + 4x = 180°$

$\Rightarrow 9x = 180°$

$\Rightarrow x = 180°/9°$

$\Rightarrow x = 20°$

Substituting the value of x in two adjacent angles

$5x = 5(20)° = 100°$

$4x = 4(20)° = 80°$

Therefore, the measure of each one of these angles is 100° and 80°

Solution 10: Let two straight lines AB and CD intersect at O and let $\angle AOC = 90°$.

Now, $\angle AOC = \angle BOD$ [Vertically opposite angles]

$\Rightarrow \angle BOD = 90°$

Also, as ∠AOC and ∠AOD form a linear pair.

$90^o + \angle AOD = 180^o$

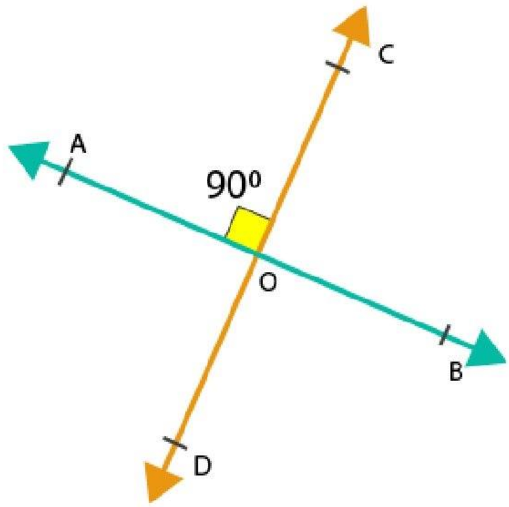
$\Rightarrow \angle AOD = 180^o - 90^o$

$\Rightarrow \angle AOD = 90^o$

Since, $\angle BOC = \angle AOD$ [Vertically opposite angles]

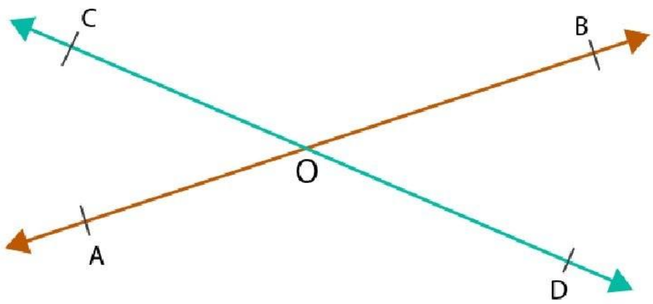
$\Rightarrow \angle BOC = 90^o$

Thus, each of the remaining angles is 90^o.



Solution 11: From the figure we know that ∠AOD and ∠BOC are vertically opposite angles

$\therefore \angle AOD = \angle BOC$



It is given that

$\angle BOC + \angle AOD = 280^o$

$\angle AOD + \angle AOD = 280^o$ [$\because \angle AOD = \angle BOC$]

$2\angle AOD = 280^o$

$\angle AOD = 280/2$

$\angle AOD = \angle BOC = 140^o$

Again, from the figure we have ∠AOC and ∠AOD form a linear pair

So,

$\angle AOC + \angle AOD = 180^o$

$\angle AOC + 140^o = 180^o$

$\angle AOC = 180^o - 140^o$

$\angle AOC = 40^o$

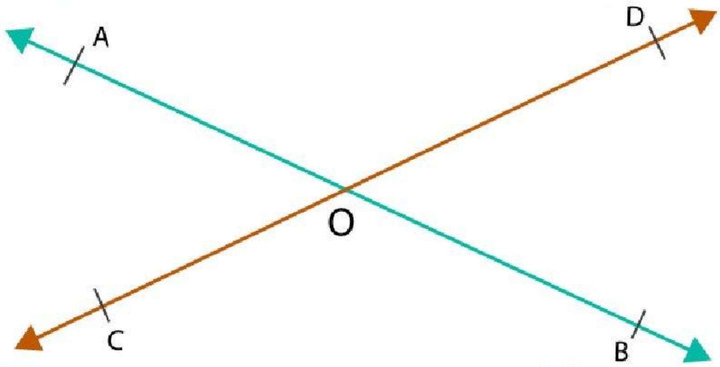
Now, from the figure we have ∠AOC and ∠BOD are vertically opposite angles

$\therefore \angle AOC = \angle BOD = 40^o$

Therefore, $\angle AOC = 40^o$, $\angle BOC = 140^o$, $\angle AOD = 140^o$ and $\angle BOD = 40^o$

Solution 12: Let $\angle AOC = 5k$ and $\angle AOD = 7k$, where k is some constant.

Here, $\angle AOC$ and $\angle AOD$ form a linear pair.



$\therefore \angle AOC + \angle AOD = 180^\circ$

$\Rightarrow 5k + 7k = 180^\circ$

$\Rightarrow 12k = 180^\circ$

$\Rightarrow k = 15^\circ$

$\therefore \angle AOC = 5k = 5 \times 15^\circ = 75^\circ$

$\angle AOD = 7k = 7 \times 15^\circ = 105^\circ$

Now, $\angle BOD = \angle AOC = 75^\circ$ (Vertically opposite angles)

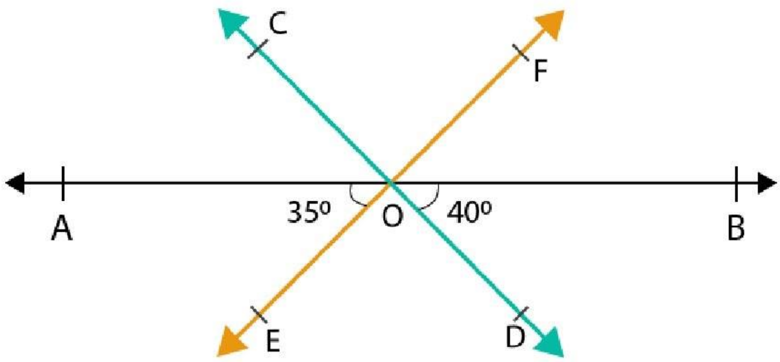
$\angle BOC = \angle AOD = 105^\circ$ (Vertically opposite angles)

Solution 13: In the given figure,

$\angle AOC = \angle BOD = 40^\circ$ (Vertically opposite angles)

$\angle BOF = \angle AOE = 35^\circ$ (Vertically opposite angles)

Now, $\angle EOC$ and $\angle COF$ form a linear pair.



$\therefore \angle EOC + \angle COF = 180^\circ$

$\Rightarrow (\angle AOE + \angle AOC) + \angle COF = 180^\circ$

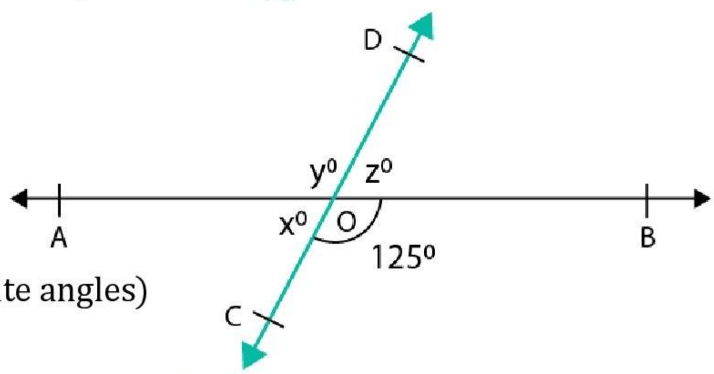
$\Rightarrow 35^\circ + 40^\circ + \angle COF = 180^\circ$

$\Rightarrow 75^\circ + \angle COF = 180^\circ$

$\Rightarrow \angle COF = 180^\circ - 75^\circ = 105^\circ$

Also, $\angle DOE = \angle COF = 105^\circ$ (Vertically opposite angles)

Solution 14: Here, $\angle AOC$ and $\angle BOC$ form a linear pair.



$\therefore \angle AOC + \angle BOC = 180^\circ$

$\Rightarrow x^\circ + 125^\circ = 180^\circ$

$\Rightarrow x^\circ = 180^\circ - 125^\circ = 55^\circ$

Now,

$\angle AOD = \angle BOC = 125^\circ$ (Vertically opposite angles)

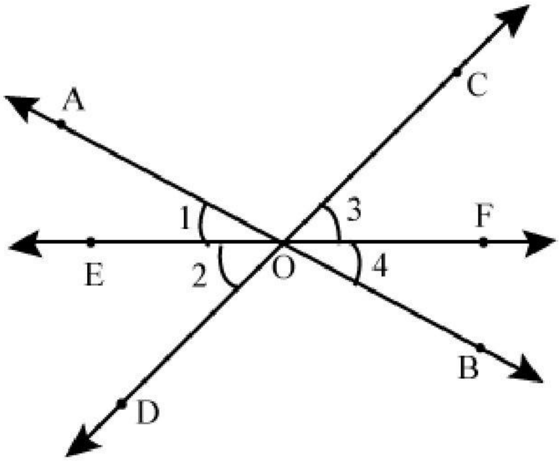
$\therefore y^\circ = 125^\circ$

$\angle BOD = \angle AOC = 55^\circ$ (Vertically opposite angles)

$\therefore z^o = 55^o$

Thus, the respective values of x, y and z are 55^o, 125^o and 55^o.

Solution 15: Let AB and CD be the two lines intersecting at a point O and let ray OE bisect $\angle AOD$.
Now, draw a ray OF in the opposite direction of OE, such that EOF is a straight line.



To prove: OF is the bisector of $\angle BOC$

Let $\angle AOE = 1$, $\angle EOD = 2$, $\angle COF = 3$ and $\angle BOF = 4$

We know that vertically-opposite angles are equal.

$\therefore \angle 1 = \angle 4$ and $\angle 2 = \angle 3$

But, $\angle 1 = \angle 2$ [Since OE bisects $\angle AOD$]

$\therefore \angle 4 = \angle 3$

Hence, the ray opposite the bisector of one of the angles so formed bisects the vertically-opposite angle

Proof: $\because \angle AOD = \angle BOC$ (Vertically opposite angles)

$\angle 1 = \angle 2$ ($\because$ OE is the bisector of $\angle AOD$)

$\angle 1 = \angle 4$ (Vertically opposite angles)

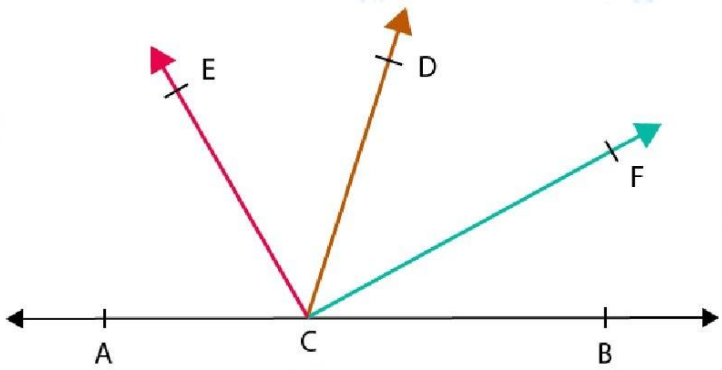
$\angle 2 = \angle 3$ (Vertically opposite angles)

But $\angle 1 = \angle 2$

$\therefore \angle 3 = \angle 4$

Hence, OF is the bisector of $\angle BOC$

Solution 16:



To prove: $\angle ECF = 90°$

Let ACB denote a straight line and let $\angle ACD$ and $\angle BCD$ be the supplementary angles.

Then, we have:

$\angle ACD = x°$ and $\angle BCD = (180 - x)°$

Let CE bisect $\angle ACD$ and CF bisect $\angle BCD$

Then, we have:

$\angle ACE = \angle ECD = \frac{1}{2}x°$ and

$\angle BCF = \angle FCD = \frac{1}{2}(180 - x)°$

Therefore,

$\angle ECD + \angle FCD = \frac{1}{2}x + \frac{1}{2}(180° - x)$

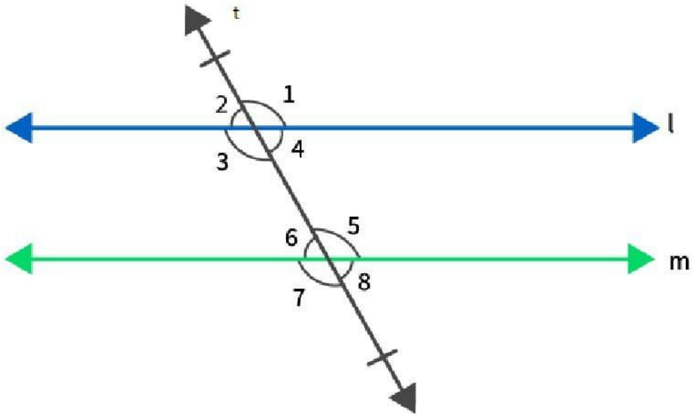
$\Rightarrow \angle ECF = \frac{1}{2}(x + 180° - x)$

$\Rightarrow \angle ECF = \frac{1}{2}(180°)$

$\Rightarrow \angle ECF = 90°$

Exercise 7C

Solution 1: We have, $\angle 1 = 120°$. Then,



$\angle 1 = \angle 5$ [Corresponding angles]

$\Rightarrow \angle 5 = 120°$

$\angle 1 = \angle 3$ [Vertically-opposite angles]

$\Rightarrow \angle 3 = 120°$

$\angle 5 = \angle 7$ [Vertically-opposite angles]

$\Rightarrow \angle 7 = 120°$

$\angle 1 + \angle 2 = 180°$ [Since AFB is a straight line]

$\Rightarrow 120° + \angle 2 = 180°$

$\Rightarrow \angle 2 = 60°$

$\angle 2 = \angle 4$ [Vertically-opposite angles]

$\Rightarrow \angle 4 = 60°$

$\angle 2 = \angle 6$ [Corresponding angles]

$\Rightarrow \angle 6 = 60°$

$\angle 6 = \angle 8$ [Vertically-opposite angles]

$\Rightarrow \angle 8 = 60°$

So the corresponding angles according to the figure is written as

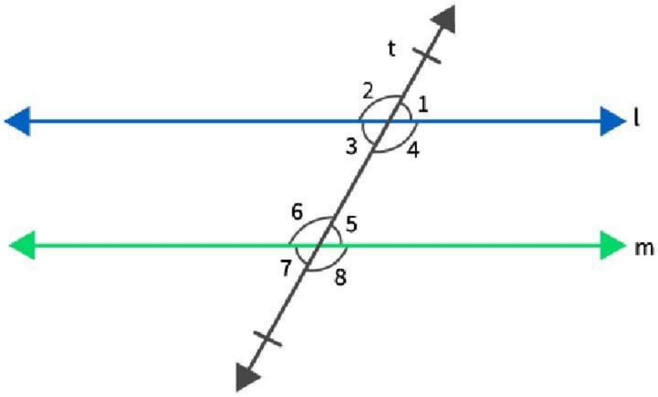
$\angle 1 = \angle 5 = 120^o$

$\angle 2 = \angle 6 = 60^o$

$\angle 3 = \angle 7 = 120^o$

$\angle 4 = \angle 8 = 60^{o}$

Solution 2: Given, $\angle 7 = 80°$



Now, $\angle 7 + \angle 8 = 180°$ (linear pair)

$\Rightarrow 80° + \angle 8 = 180°$

$\Rightarrow \angle 8 = 100°$

$\angle 7 = \angle 5$ (vertically opposite angles)

$\Rightarrow \angle 5 = 80°$

Also, $\angle 6 = \angle 8$ (vertically opposite angles)

$\Rightarrow \angle 6 = 100°$

Line l || line m and line t is a transversal.

$\Rightarrow \angle 1 = \angle 5 = 80°$ (corresponding angles)

$\angle 2 = \angle 6 = 100°$ (corresponding angles)

$\angle 3 = \angle 7 = 80°$ (corresponding angles)

$\angle 4 = \angle 8 = 100°$ (corresponding angles)

Solution 3: Given, $\angle 1 : \angle 2 = 2 : 3$

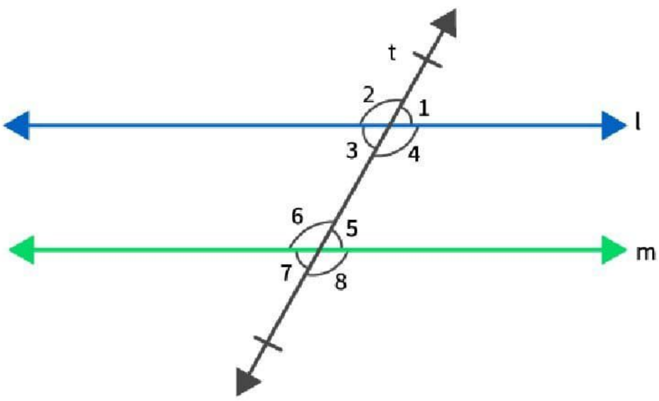
Now, $\angle 1 + \angle 2 = 180°$ (linear pair)

$\Rightarrow 2x + 3x = 180°$

$\Rightarrow 5x = 180°$

$\Rightarrow x = 36°$

$\Rightarrow \angle 1 = 2x = 72°$ and $\angle 2 = 3x = 108°$



$\angle 1 = \angle 3$ (vertically opposite angles)

$\Rightarrow \angle 3 = 72°$

Also, $\angle 2 = \angle 4$ (vertically opposite angles)

$\Rightarrow \angle 4 = 108°$

Line l || line m and line t is a transversal.

$\angle 5 = \angle 1 = 72°$ (corresponding angles)

$\angle 6 = \angle 2 = 108°$ (corresponding angles)

$\angle 7 = \angle 3 = 72°$ (corresponding angles)

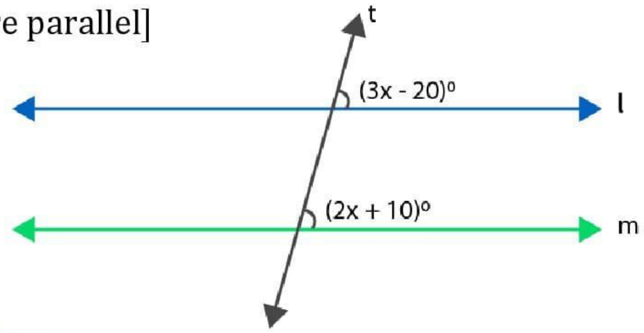
$\angle 8 = \angle 4 = 108°$ (corresponding angles)

Solution 4: Lines l and m will be parallel if 3x - 20 = 2x + 10

[Since, if corresponding angles are equal, lines are parallel]

$\Rightarrow 3x - 2x = 10 + 20$

$\Rightarrow x = 30$



Solution 5: We know that both the angles are consecutive interior angles

So, it can be written as

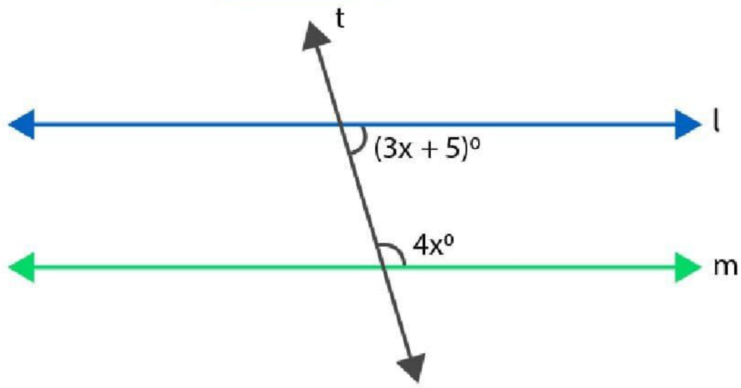
$3x + 5 + 4x = 180$

$\Rightarrow 7x = 180 – 5$

$\Rightarrow 7x = 175$

$\Rightarrow x = 25$

Therefore, the value of x is 25.



Solution 6:

Since AB || CD and BC is a transversal.

So, $\angle BCD = \angle ABC = x°$ [Alternate angles]

As BC || ED and CD is a transversal.

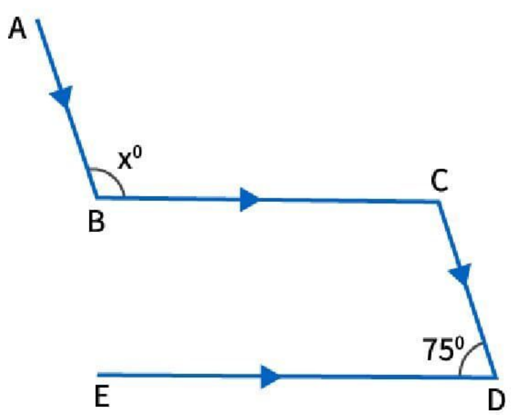
$\angle BCD + \angle EDC = 180°$

$\Rightarrow \angle BCD + 75° = 180°$

$\Rightarrow \angle BCD = 180° - 75°$

$\Rightarrow \angle BCD = 105°$

$\Rightarrow \angle ABC = 105°$ [since $\angle BCD = \angle ABC$]



$\therefore x^{\circ} = ABC = 105^{\circ}$

Hence, $x = 105$.

Solution 7: Since AB || CD and BC is a transversal.

So, $\angle ABC = \angle BCD$ [alternate interior angles]

$\Rightarrow 70^{\circ} = x^{\circ} + \angle ECD$ ...(i)

Now, CD || EF and CE is transversal.

So, $\angle ECD + \angle CEF = 180^{\circ}$ [sum of consecutive interior angles is 180°]

$\therefore \angle ECD + 130^{\circ} = 180^{\circ}$

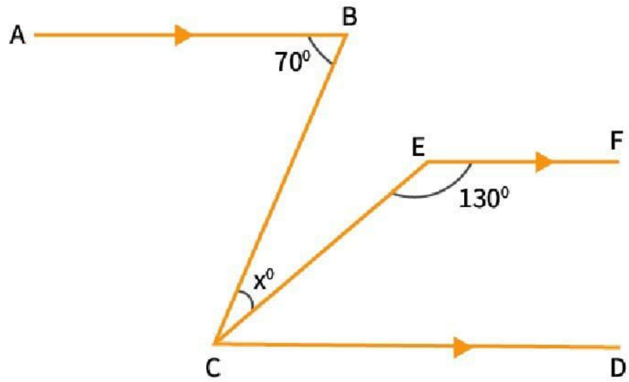
$\Rightarrow \angle ECD = 180^{\circ} - 130^{\circ} = 50^{\circ}$

Putting $\angle ECD = 50^{\circ}$ in (i) we get,

$70^{\circ} = x^{\circ} + 50^{\circ}$

$\Rightarrow x = 70^{\circ} - 50^{\circ}$

$\Rightarrow x = 20^{\circ}$

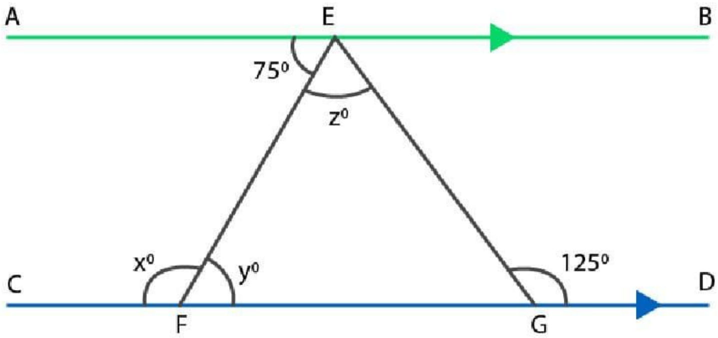


Solution 8: AB || CD and EF is transversal.

$\Rightarrow \angle AEF = \angle EFG$ (alternate angles)

Given, $\angle AEF = 75°$

$\Rightarrow \angle EFG = y = 75°$



Now, $\angle EFC + \angle EFG = 180°$ (linear pair)

$\Rightarrow x + y = 180°$

$\Rightarrow x + 75° = 180°$

$\Rightarrow x = 105°$

$\angle EGD = \angle EFG + \angle FEG$ (Exterior angle property)

$\Rightarrow 125° = y + z$

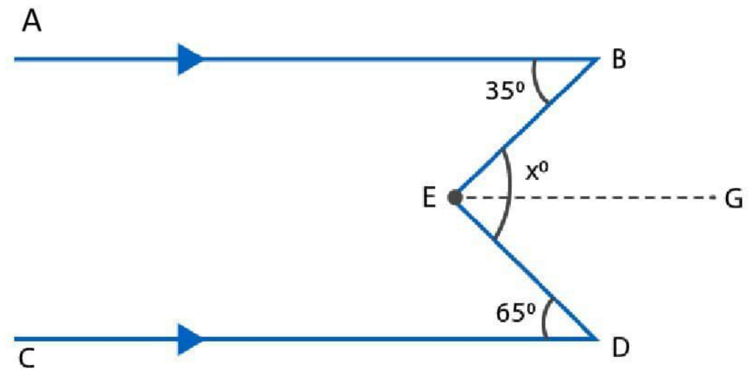
$\Rightarrow 125° = 75° + z$

$\Rightarrow z = 50°$

Thus, $x = 105°$, $y = 75°$ and $z = 50°$

Solution 9:

(i) Through E draw EG || CD. Now since EG||CD and ED is a transversal.



So, $\angle GED = \angle EDC = 65^o$ [Alternate interior angles]

Since EG || CD and AB || CD,

EG||AB and EB is transversal.

So, $\angle BEG = \angle ABE = 35^o$[Alternate interior angles]

So, $\angle DEB = x^o$

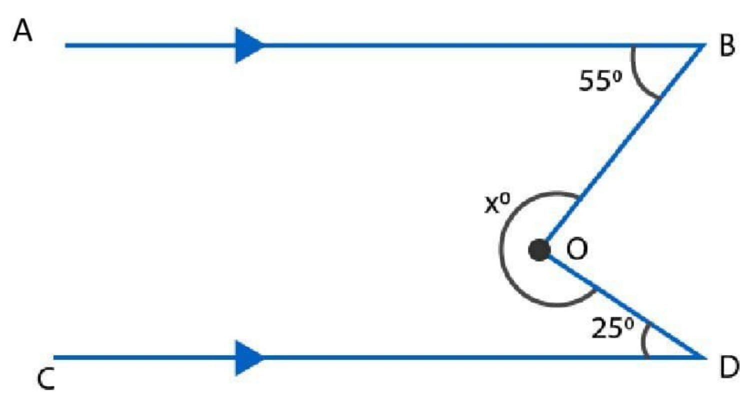
$\Rightarrow \angle BEG + \angle GED = x^o$

$\Rightarrow 35^o + 65^o = x^o$

$\Rightarrow 100^o = x^o$

Hence, $x = 100$.

(ii)



Now since OF || CD and OD is transversal.

$\angle CDO + \angle FOD = 180^o$

[sum of consecutive interior angles is 180^o]

$\Rightarrow 25^o + \angle FOD = 180^o$

$\Rightarrow \angle FOD = 180^o - 25^o = 155^o$

As OF || CD and AB || CD [Given]

Thus, OF || AB and OB is a transversal.

So, $\angle ABO + \angle FOB = 180^o$ [sum of consecutive interior angles is 180^o]

$\Rightarrow 55^o + \angle FOB = 180^o$

$\Rightarrow \angle FOB = 180^o - 55^o$

$\Rightarrow \angle FOB = 125^o$

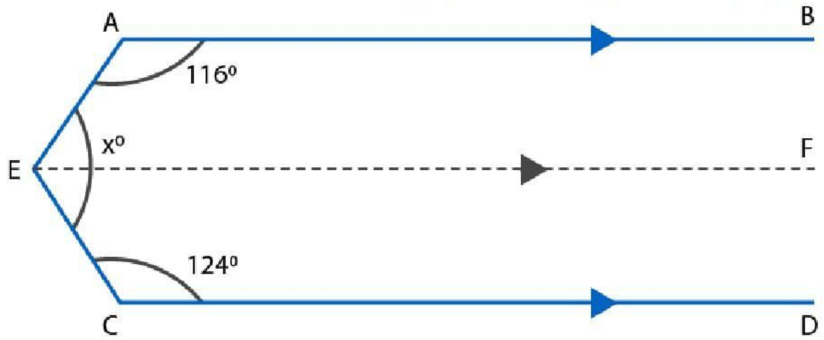
Now, $x^o = \angle FOB + \angle FOD$

$= 125^o + 155^o$

$= 280^o$.

Hence, x = 280.

(iii) Through E, draw EF || CD.



Now since EF || CD and EC is transversal.

$\angle FEC + \angle ECD = 180^o$

[sum of consecutive interior angles is 180^o]

$\Rightarrow \angle FEC + 124^o = 180^o$

$\Rightarrow \angle FEC = 180^o - 124^o$

$\Rightarrow \angle FEC = 56^o$

Since EF || CD and AB ||CD

So, EF || AB and AE is a transversal.

So, $\angle BAE + \angle FEA = 180^{\circ}$

[sum of consecutive interior angles is 180°]

$\therefore 116^{\circ} + \angle FEA = 180^{\circ}$

$\Rightarrow \angle FEA = 180^{\circ} - 116^{\circ}$

$\Rightarrow \angle FEA = 64^{\circ}$

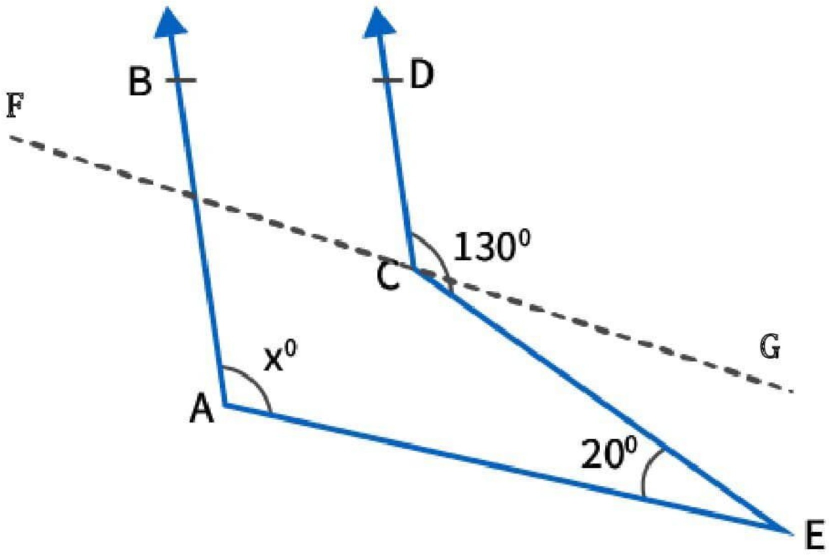
Thus, $x^{\circ} = \angle FEA + \angle FEC$

$= 64^{\circ} + 56^{\circ}$

$= 120^{\circ}$.

Hence, $x = 120^{\circ}$

Solution 10: Draw a line through point C and name it as FG where FG || AE



Here FG || AE and CE is a transversal

From the figure we have $\angle GCE$ and $\angle CEA$ are alternate angles

So,

$\angle GCE = \angle CEA = 20^{\circ}$

$\angle DCG = \angle DCE - \angle GCE$

$\angle DCG = 130^{\circ} - 20^{\circ}$

$\angle DCG = 110^{\circ}$

Again AB || CD and FG is a transversal

From the figure we have ∠BFC and ∠DCG are corresponding angles

So,

$\angle BFC = \angle DCG = 110^o$

Also, FG || AE and AF is a transversal

From the figure we know that ∠BFG and ∠FAE are corresponding angles

So,

$\angle BFG = \angle FAE = 110^o$

$\angle FAE = x = 110^o$

Therefore, the value of x is 110^o.

Solution 11: It is given that AB || PQ and EF is a transversal

So,

$\angle CEB = \angle EFQ = 75^o$ [corresponding angles]

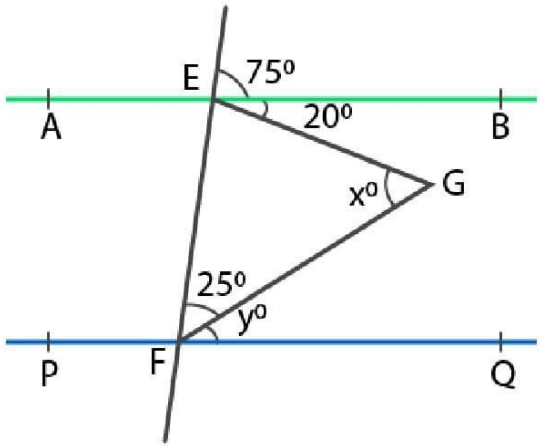
$\Rightarrow \angle EFQ = 75^o$

$\Rightarrow \angle EFG + \angle GFQ = 75^o$

$\Rightarrow 25^o + y^o = 75^o$

$\Rightarrow y^o = 75^o - 25^o$

$\Rightarrow y^o = 50^o$



Also,

$\angle BEF + \angle EFQ = 180^o$ [sum of consecutive interior angles]

$\Rightarrow \angle BEF + 75^o = 180^o$

$\Rightarrow \angle BEF = 180^o - 75^o$

$\Rightarrow \angle BEF = 105^o$

Now, ∠BEF can be written as

$\angle BEF = \angle FEG + \angle GEB$

$\Rightarrow 105^o = \angle FEG + 20^o$

$\Rightarrow \angle FEG = 105^\circ - 20^\circ$

$\Rightarrow \angle FEG = 85^\circ$

In $\triangle$ EFG we have,

$x^\circ + 25^\circ + \angle FEG = 180^\circ$

$\Rightarrow x^\circ + 25^\circ + 85^\circ = 180^\circ$

$\Rightarrow x^\circ = 180^\circ - 25^\circ - 85^\circ$

$\Rightarrow x^\circ = 180^\circ - 110^\circ$

$\Rightarrow x^\circ = 70^\circ$

Therefore, the value of x is 70.

Solution 12: It is given that AB || CD and AC is a transversal.

So,

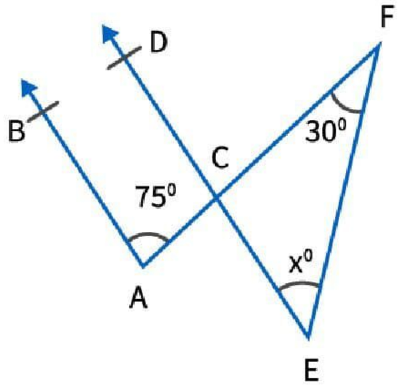
$\angle BAC + \angle ACD = 180^\circ$ [sum of consecutive interior angles is 180°]

$\Rightarrow 75^\circ + \angle ACD = 180^\circ$

$\Rightarrow \angle ACD = 180^\circ - 75^\circ$

$\Rightarrow \angle ACD = 105^\circ$

$\angle ECF = \angle ACD = 105^\circ$ [vertically opposite angles]



In $\triangle$ CEF we have,

$\angle ECF + \angle CEF + \angle EFC = 180^\circ$

$\Rightarrow 105^\circ + x^\circ + 30^\circ = 180^\circ$

$\Rightarrow x^\circ = 180^\circ - 105^\circ - 30^\circ$

$\Rightarrow x^\circ = 180^\circ - 135^\circ$

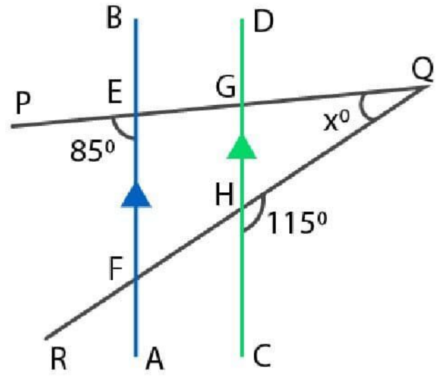
$\Rightarrow x^\circ = 45^\circ$

Therefore, the value of x is 45.

Solution 13: It is given that AB || CD and PQ is a transversal

So,

$\angle PEF = \angle EGH = 85^o$ [corresponding angles]



Now,

$\angle EGH + \angle QGH = 180^o$ [linear pair angle]

$\Rightarrow 85^o + \angle QGH = 180^o$

$\Rightarrow \angle QGH = 180^o - 85^o$

$\Rightarrow \angle QGH = 95^o$

Similarly,

$\Rightarrow \angle GHQ + \angle CHQ = 180^o$ [linear pair angle]

$\Rightarrow \angle GHQ + 115^o = 180^o$

$\Rightarrow \angle GHQ = 180^o - 115^o$

$\Rightarrow \angle GHQ = 65^o$

In $\triangle$ GHQ we get,

$\angle GQH + \angle GHQ + \angle QGH = 180^o$

By substituting the values

$x^o + 65^o + 95^o = 180^o$

$x^o = 180^o - 65^o - 95^o$

$x^o = 180^o - 160^o$

$x^o = 20^o$

Therefore, the value of x is 20^o

Solution 14: It is given that AB || CD and BC is a transversal

From the figure we have

$\angle ABC = \angle BCD$

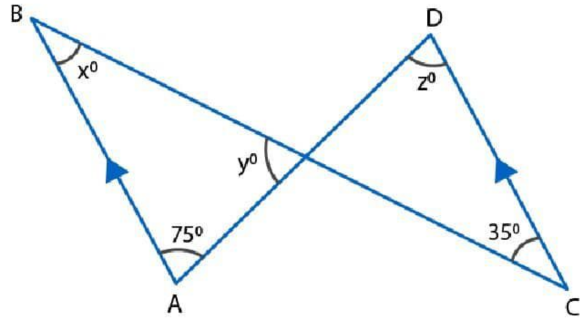
$x = 35$

It is also given that AB || CD and AD is a transversal

So,

$\angle BAD = \angle ADC$

$z = 75$



In $\triangle$ ABO we have,

$\angle ABO + \angle BAO + \angle BOA = 180^o$

$\Rightarrow x^o + 75^o + y^o = 180^o$

$\Rightarrow 35^o + 75^o + y^o = 180^o$

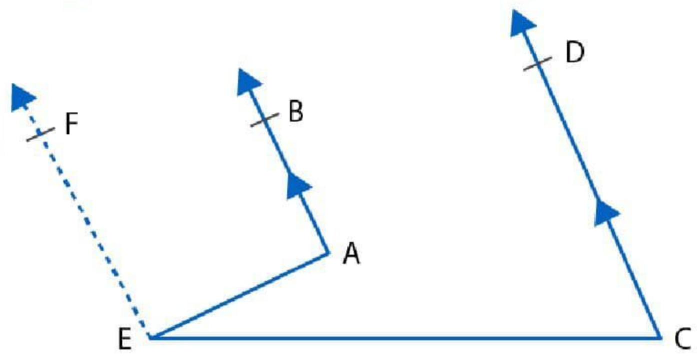
$\Rightarrow y^o = 180^o - 35^o - 75^o$

$\Rightarrow y^o = 180^o - 110^o$

$\Rightarrow y^o = 70^o$

Therefore, the value of x, y and z is 35, 70 and 75.

Solution 15: Draw a line EF which is parallel to AB and CD through the point E



Draw EF || AB || CD through E.

Now, EF || AB and AE is the transversal.

Then,

$\angle BAE + \angle AEF = 180°$ [Angles on the same side of a transversal line are supplementary]

Again, EF | |CD and CE is the transversal.

Then,

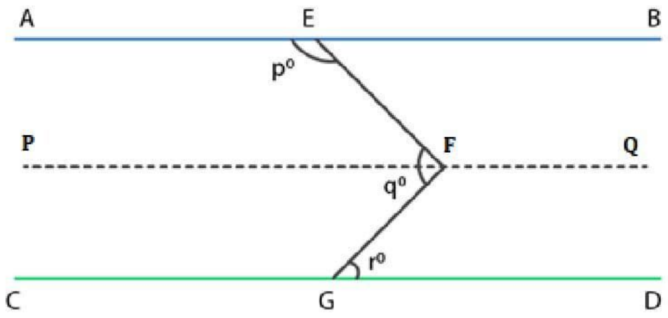
$\angle DCE + \angle CEF = 180°$ [Angles on the same side of a transversal line are supplementary]

$\Rightarrow \angle DCE + (\angle AEC + \angle AEF) = 180°$

$\Rightarrow \angle DCE + \angle AEC + 180° - \angle BAE = 180°$

$\Rightarrow \angle BAE - \angle DCE = \angle AEC$

Solution 16:



Draw PFQ || AB || CD

Now, PFQ || AB and EF is the transversal.

Then,

$\angle AEF + \angle EFP = 180°$ .....(i)

[Angles on the same side of a transversal line are supplementary]

Also, PFQ || CD

$\angle PFG = \angle FGD = r$ [Alternate angles]

and $\angle EFP = \angle EFG - \angle PFG = q - r$

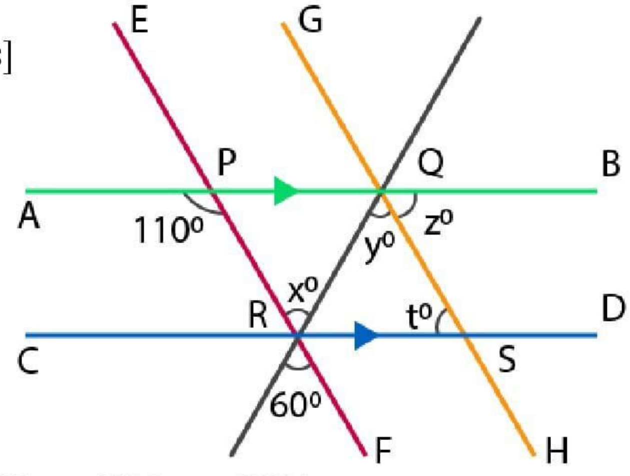
$\angle AEF = p°$

putting the value of $\angle EFP$ and $\angle AEF$ in eqn. (i)we get,

$p + q - r = 180°$

Hence proved.

Solution 17:



$\angle PRQ = x° = 60°$ [vertically opposite angles]

Since EF || GH, and RQ is a transversal.

So, $\angle x = \angle y$ [Alternate angles]

$\Rightarrow \angle y = 60°$

AB || CD and PR is a transversal.

So, $\angle PRD = \angle APR$ [Alternate angles]

$\Rightarrow \angle PRQ + \angle QRD = \angle APR$ [since $\angle PRD = \angle PRQ + \angle QRD$]

$\Rightarrow x + \angle QRD = 110°$

$\Rightarrow \angle QRD = 110^o - 60^o = 50^o$

In ΔQRS, we have,

$\angle QRD + t^o + y^o = 180^o$

$\Rightarrow 50^o + t + 60 = 180^o$

$\Rightarrow t = 180^o - 110^o$

$\Rightarrow t = 70^o$

Since, AB || CD and GH is a transversal

So, $z^o = t^o = 70^o$ [Alternate angles]

$\therefore x = 60^o, y = 60^o, z = 70^o$ and $t = 70^o$

Solution 18: Here AB || CD and t is a transversal cutting at points E and F

So,

$\angle BEF + \angle DFE = 180^o$ [interior angles]

$\Rightarrow \frac{1}{2}\angle BEF + \frac{1}{2}\angle DFE = 90°$

$\Rightarrow \angle GEF + \angle GFE = 90^o$ .... (1)

In $\triangle$ GEF we have,

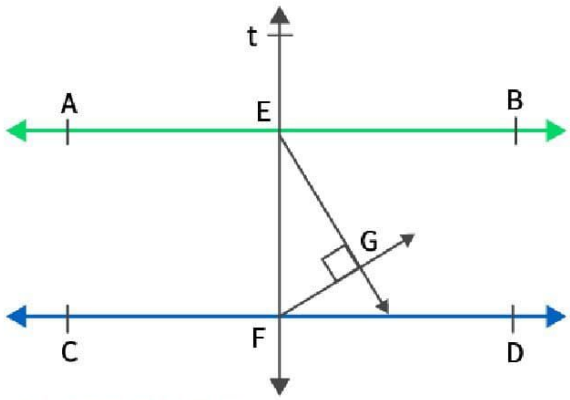
$\angle GEF + \angle GFE + \angle EGF = 180^o$

$90^o + \angle EGF = 180^o$

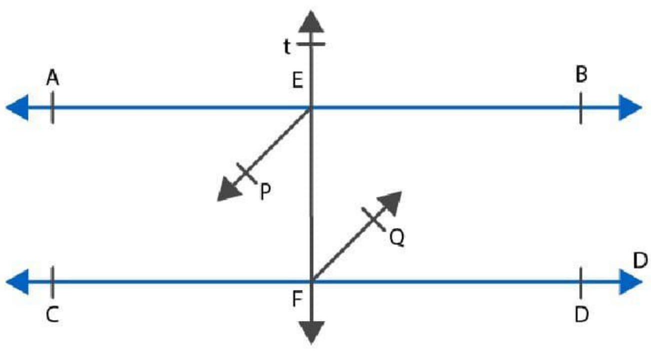
$\Rightarrow \angle EGF = 180^o - 90^o$ [From(i)]

$\Rightarrow \angle EGF = 90^o$

Therefore, it is proved that $\angle EGF = 90^o$



Solution 19:



Since AB ∥ CD and t is a transversal, we have

$\angle AEF = \angle EFD$ (alternate angles)

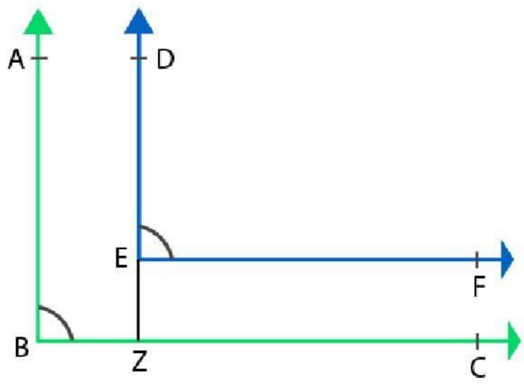
$\Rightarrow \frac{1}{2}\angle AEF = \frac{1}{2}\angle EFD$

$\Rightarrow \angle PEF = \angle EFQ$

But, these are alternate interior angles formed when the transversal EF cuts EP and FQ.

$\therefore$ EP ∥ FQ

Solution 20: Construction: Produce DE to meet BC at Z.



Now, AB ∥ DZ and BC is the transversal.

$\Rightarrow \angle ABC = \angle DZC$ (corresponding angles) ....(i)

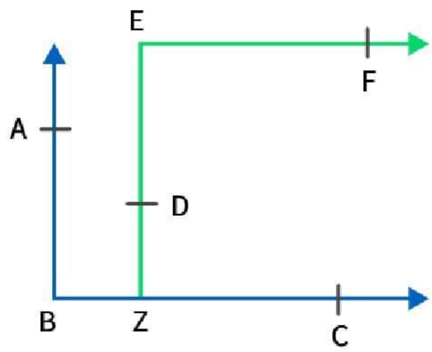
Also, EF ∥ BC and DZ is the transversal.

$\Rightarrow \angle DZC = \angle DEF$ (corresponding angles) ....(ii)

From (i) and (ii), we have

$\angle ABC = \angle DEF$

Solution 21: Construction: Produce ED to meet BC at Z.



Now, AB ∥ EZ and BC is the transversal.

$\Rightarrow \angle ABZ + \angle EZB = 180°$ (interior angles)

$\Rightarrow \angle ABC + \angle EZB = 180°$ ....(i)

Also, EF ∥ BC and EZ is the transversal.

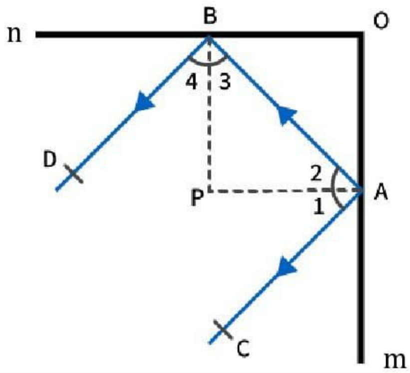
$\Rightarrow \angle BZE = \angle ZEF$ (alternate angles)

$\Rightarrow \angle BZE = \angle DEF$ ....(ii)

From (i) and (ii), we have

$\angle ABC + \angle DEF = 180°$

Solution 22:



Let the normal to mirrors m and n intersect at O.

Now, $OB \perp m$, $OA \perp n$ and $m \perp n$.

$\Rightarrow OB \perp OA$

$\Rightarrow \angle AOB = 90°$

$\Rightarrow \angle 2 + \angle 3 = 90°$ (sum of acute angles of a right triangle is 90°)

By the laws of reflection, we have

$\angle 1 = \angle 2$ and $\angle 4 = \angle 3$ (angle of incidence = angle of reflection)

$\Rightarrow \angle 1 + \angle 4 = \angle 2 + \angle 3 = 90°$

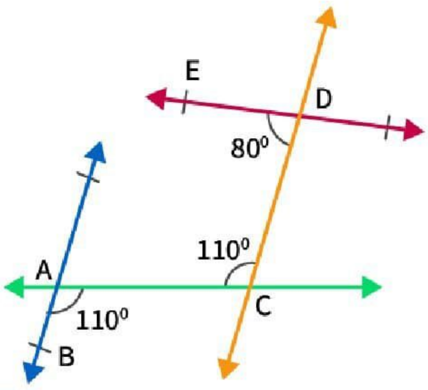
$\therefore \angle 1 + \angle 2 + \angle 3 + \angle 4 = 90° + 90°$

$\Rightarrow \angle CAB + \angle ABD = 180°$

But, $\angle CAB$ and $\angle ABD$ are consecutive interior angles formed, when the transversal AB cuts CA and BD.

$\therefore$ CA || BD

Solution 23: In the given figure,

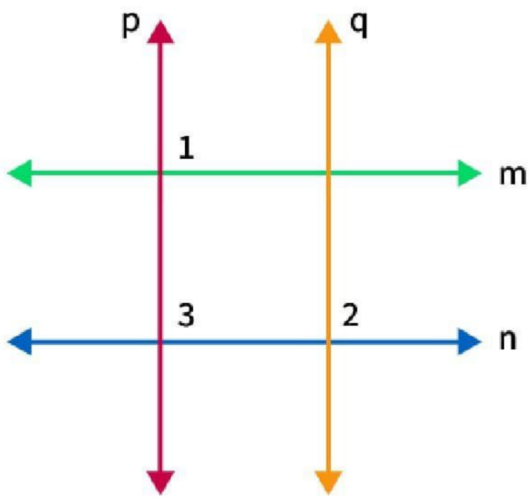


$\angle BAC = \angle ACD = 110°$

But, these are alternate angles when transversal AC cuts AB and CD.

Hence, AB || CD.

Solution 24:



Let the two parallel lines be m and n.

Let $p \perp m$.

$\Rightarrow \angle 1 = 90°$

Let $q \perp n$.

$\Rightarrow \angle 2 = 90°$

Now, $m \parallel n$ and p is a transversal.

$\Rightarrow \angle 1 = \angle 3$ (corresponding angles)

$\Rightarrow \angle 3 = 90°$

$\Rightarrow \angle 3 = \angle 2$ (each 90°)

But, these are corresponding angles, when transversal n cuts lines p and q.

$\therefore p \parallel q$.

Hence, two lines which are perpendicular to two parallel lines, are parallel to each other.

Multiple Choice Questions (MCQ)

Solution 1: (d) a right triangle

Let ΔABC be such that $\angle A = \angle B + \angle C$.

In ΔABC,

$\angle A + \angle B + \angle C = 180^\circ$ (Angle sum property)

$\Rightarrow \angle A + \angle A = 180^\circ$ ($\angle A = \angle B + \angle C$)

$\Rightarrow 2\angle A = 180^\circ$

$\Rightarrow \angle A = 90^\circ$

Therefore, ΔABC is a right triangle.

Thus, if one angle of a triangle is equal to the sum of the other two angles, then the triangle is a right triangle.

Hence, the correct answer is option (d).

Solution 2: (b) 55°

Let the measure of each of the two equal interior opposite angles of the triangle be x.

In a triangle, the exterior angle is equal to the sum of the two interior opposite angles.

$\therefore x + x = 110^\circ$

$\Rightarrow 2x = 110^\circ$

$\Rightarrow x = 55^\circ$

Thus, the measure of each of these equal angles is 55°.

Hence, the correct answer is option (b).

Solution 3: (a) acute-angled

Let the angles measure $(3x)^\circ$, $(5x)^\circ$ and $(7x)^\circ$

Then,

$3x + 5x + 7x = 180^\circ$

$\Rightarrow 15x = 180°$

$\Rightarrow x = 12°$

Therefore, the angles are

$(3x)° = 3(12)° = 36°$

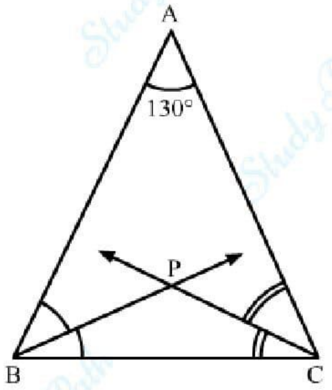
$(5x)° = 5(12)° = 60°$ and

$(7x)° = 7(12)° = 84°$

Hence, the triangle is acute-angled.

Solution 4: (d) 155°

Let ΔABC be such that $\angle A = 130°$.



Here, BP is the bisector of $\angle B$ and CP is the bisector of $\angle C$.

$\therefore \angle ABP = \angle PBC = \frac{1}{2}\angle B \quad .....(1)$

Also, $\angle ACP = \angle PCB = \frac{1}{2}C \quad .....(2)$

In ΔABC,

$\angle A + \angle B + \angle C = 180°$ (Angle sum property)

$\Rightarrow 130° + \angle B + \angle C = 180°$

$\Rightarrow \angle B + \angle C = 180° - 130° = 50°$

$\Rightarrow \frac{1}{2}\angle B + \frac{1}{2}\angle C = \frac{1}{2} \times 50° = 25°$

$\Rightarrow \angle PBC + \angle PCB = 25°$ .....(3) [Using (1) and (2)]

In ΔPBC,

$\angle PBC + \angle PCB + \angle BPC = 180°$ (Angle sum property)

$\Rightarrow 25° + \angle BPC = 180°$ [Using (3)]

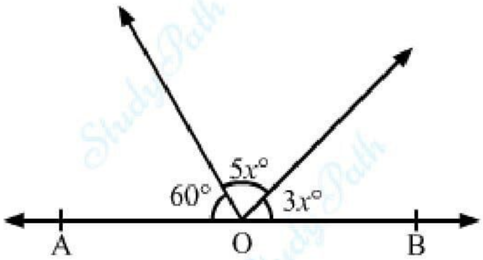
$\Rightarrow \angle BPC = 180° - 25° = 155°$

Thus, if one of the angles of a triangle is 130° then the angle between the bisectors of the other two angles is 155°.

Hence, the correct answer is option (d).

Solution 5: (b) 15

It is given that, AOB is a straight line.



$\therefore 60° + (5x° + 3x°) = 180°$ (Linear pair)

$\Rightarrow 8x° = 180° - 60°$

$\Rightarrow 8x° = 120°$

$\Rightarrow x° = 15°$

Thus, the value of x is 15.

Hence, the correct answer is option (b).

Solution 6: (c) 80°

Suppose ΔABC be such that $\angle A : \angle B : \angle C = 2 : 3 : 4$.

Let $\angle A = 2k$, $\angle B = 3k$ and $\angle C = 4k$, where k is some constant.

In ΔABC,

$\angle A + \angle B + \angle C = 180°$ (Angle sum property)

$\Rightarrow 2k + 3k + 4k = 180°$

$\Rightarrow 9k = 180$

$\Rightarrow k = 20°$

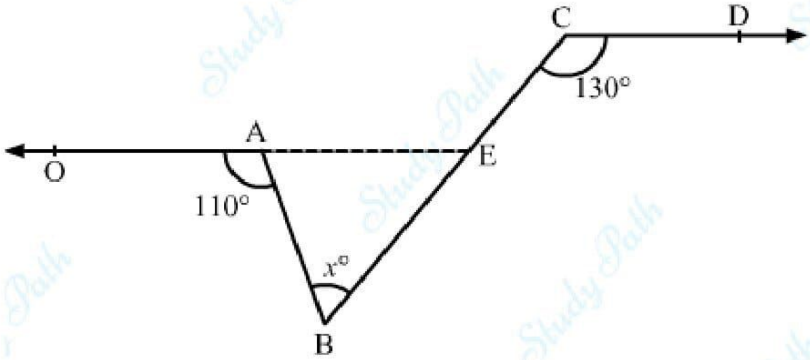
$\therefore$ Measure of the largest angle $= 4k = 4 \times 20° = 80°$

Hence, the correct answer is option (c).

Solution 7: (c) 60°

In the given figure, OA || CD.

Construction: Extend OA such that it intersects BC at E.



Now, OE || CD and BC is a transversal.

$\therefore \angle AEC = \angle BCD = 130°$ (Pair of corresponding angles)

Also, $\angle OAB + \angle BAE = 180°$ (Linear pair)

$\therefore 110° + \angle BAE = 180°$

$\Rightarrow \angle BAE = 180° - 110° = 70°$

In ΔABE,

$\angle AEC = \angle BAE + \angle ABE$ (In a triangle, exterior angle is equal to the sum of two opposite interior angles)

$\therefore 130° = 70° + x°$

$\Rightarrow x° = 130° - 70° = 60°$

Thus, the measure of angle $\angle ABC$ is 60°.

Hence, the correct answer is option (c).

Solution 8: (a) an acute angle

If two angles are complements of each other, that is, the sum of their measures is 90°, then each angle is an acute angle.

Solution 9: (d) a reflex angle

Solution 10: (d) 75°

Let the measure of the required angle be x°.

Then, the measure of its complement will be $(90-x)°$

$\therefore x = 5(90 - x)$

$\Rightarrow x = 450 - 5x$

$\Rightarrow 6x = 450$

$\Rightarrow x = 75$

Solution 11: (b) 54°

Let the measure of the required angle be x°.

Then, the measure of its complement will be$(90-x)°$

$\therefore 2x = 3(90-x)$

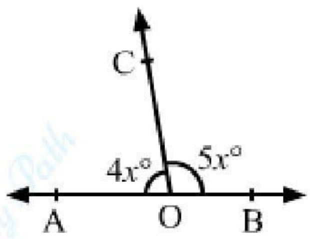
$\Rightarrow 2x = 270 - 3x$

$\Rightarrow 5x = 270$

$\Rightarrow x = 54$

Solution 12: (c) 80°

We have:



$\angle AOC + \angle BOC = 180°$ [Since AOB is a straight line]

$\Rightarrow 4x + 5x = 180°$

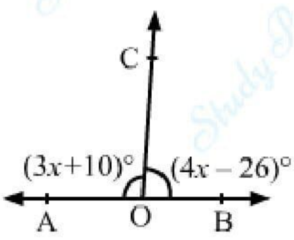
$\Rightarrow 9x = 180°$

$\Rightarrow x = 20°$

$\therefore \angle AOC = 4 \times 20° = 80°$

Solution 13: (b) 86°

We have:



$\angle AOC + \angle BOC = 180°$ [Since AOB is a straight line]

$\Rightarrow 3x + 10 + 4x - 26 = 180°$

$\Rightarrow 7x = 196°$

$\Rightarrow x = 28°$

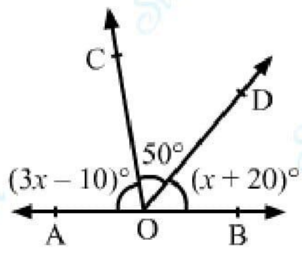
$\therefore \angle BOC = [4 \times 28 - 26]°$

Hence, $\angle BOC = 86°$.

Solution 14: (c) 80°

We have:

$\angle AOC + \angle COD + \angle BOD = 180°$ [Since AOB is a straight line]



$\Rightarrow 3x - 10 + 50 + x + 20° = 180°$

$\Rightarrow 4x = 120°$

$\Rightarrow x = 30°$

$\therefore \angle AOC = [3 \times 30 - 10]°$

$\Rightarrow \angle AOC = 80°$

Solution 15: (a) Through a given point, only one straight line can be drawn.

Clearly, statement (a) is false because we can draw infinitely many straight lines through a given point.

Solution 16: (b) 30°

Let the measure of the required angle be x°

Then, the measure of its supplement will be (180−x)°

$\therefore x = \frac{1}{5}(180° - x)$

$\Rightarrow 5x = 180° - x$

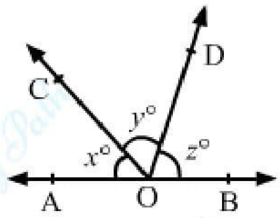
$\Rightarrow 6x = 180°$

$\Rightarrow x = 30°$

Solution 17: (a) 60°

Let

$\angle AOB = x° = (4a)°, \angle COB = y° = (5a)°$ and $\angle BOD = z° = (6a)°$



Then, we have:

$\angle AOB + \angle COB + \angle BOD = 180°$ [Since AOB is a straight line]

$\Rightarrow 4a + 5a + 6a = 180°$

$\Rightarrow 15a = 180°$

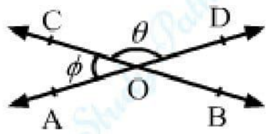
$\Rightarrow a = 12°$

Solution 18: (c) 45°

We have:

$\theta + \phi = 180°$ [$\because$AOD is a straight line]

$\Rightarrow 3\phi + \phi = 180°$ [$\because \theta = 3\phi$]



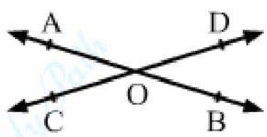
$\Rightarrow 4\phi = 180°$

$\Rightarrow \phi = 45°$

Solution 19: (b) 115°

We have :

$\angle AOC = \angle BOD$ [Vertically-Opposite Angles]



$\therefore \angle AOC + \angle BOD = 130°$

$\Rightarrow \angle AOC + \angle AOC = 130°$ [$\because \angle AOC = \angle BOD$]

$\Rightarrow 2\angle AOC = 130°$

$\Rightarrow \angle AOC = 65°$

Now,

$\angle AOC + \angle AOD = 180°$ [$\because$COD is a straight line]

$\Rightarrow 65° + \angle AOD = 180°$

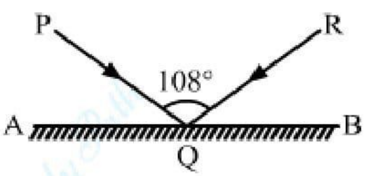
$\Rightarrow \angle AOD = 115°$

Solution 20:

(c) 36°

We know that angle of incidence = angle of reflection.

Then, let $\angle AQP = \angle BQR = x°$



Now,

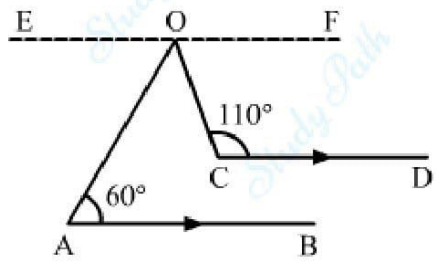
$\angle AQP + \angle PQR + \angle BQR = 180°$ [∵AQB is a straight line]

$\Rightarrow x + 108 + x = 180°$

$\Rightarrow 2x = 72°$

$\Rightarrow x = 36°$

Solution 21: (c) 50°



Draw EOF || AB || CD

Now, EO || AB and OA is the transversal.

$\therefore \angle EOA = \angle OAB = 60°$ [Alternate Interior Angles]

Also,

OF || CD and OC is the transversal.

$\therefore \angle COF + \angle OCD = 180°$ [Angles on the same side of a transversal line are supplementary]

$\Rightarrow \angle COF + 110° = 180°$

$\Rightarrow \angle COF = 70°$

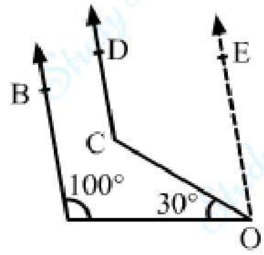
Now,

$\angle EOA + \angle AOC + \angle COF = 180°$ [∵EOF is a straight line]

$\Rightarrow 60° + \angle AOC + 70° = 180°$

$\Rightarrow \angle AOC = 50°$

Solution 22: (a) 130°



Draw OE || AB || CD

Now, OE || AB and OA is the transversal.

$\therefore \angle OAB + \angle AOE = 180°$ [Angles on the same side of a transversal line are supplementary]

$\Rightarrow \angle OAB + \angle AOC + \angle COE = 180°$

$\Rightarrow 100° + 30° + \angle COE = 180°$

$\Rightarrow \angle COE = 50°$

Also,

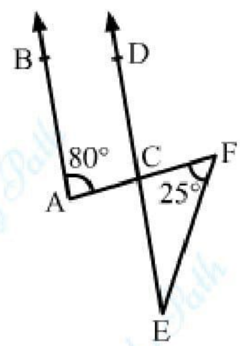
OE || CD and OC is the transversal.

$\therefore \angle OCD + \angle COE = 180°$ [Angles on the same side of a transversal line are supplementary]

$\Rightarrow \angle OCD + 50° = 180°$

$\Rightarrow \angle OCD = 130°$

Solution 23: (b) 55°



AB || CD and AF is the transversal.

$\therefore \angle DCF = \angle CAB = 80°$ [Corresponding Angles]

Side EC of triangle EFC is produced to D.

$\therefore \angle CEF + \angle EFC = \angle DCF$

$\Rightarrow \angle CEF + 25° = 80°$

$\Rightarrow \angle CEF = 55°$

Solution 24: (b) 126°

Let y=(3a)° and z=(7a)°

Let the transversal intersect AB at P, CD at O and EF at Q.

Then, we have:

$\angle COP = \angle DOF = y$ [Vertically-Opposite Angles]

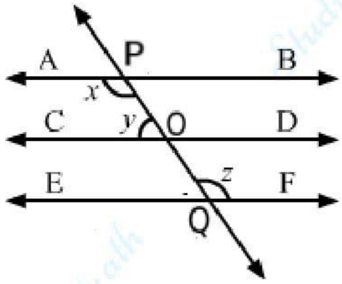
$\therefore \angle OQF + \angle DOQ = 180°$ [Consecutive Interior Angles]

$\Rightarrow 3a + 7a = 180°$

$\Rightarrow 10a = 180°$

$\Rightarrow a = 18°$

$\therefore y = 3 \times 18° = 54°$



Also,

$\angle APO + \angle COP = 180°$

$\Rightarrow x + 54° = 180°$

$\Rightarrow x = 126°$

Solution 25: (a) 50°

AB || CD and PQ is the transversal.

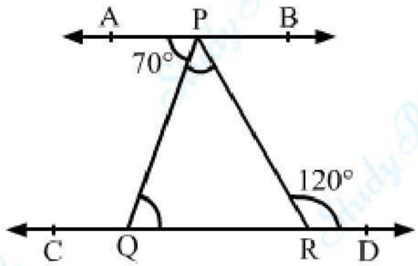
$\therefore \angle PQR = \angle APQ = 70°$ [Alternate Interior Angles]

Side QR of triangle PQR is produced to D.

$\therefore \angle PQR + \angle QPR = \angle PRD$

$\Rightarrow 70° + \angle QPR = 120°$

$\Rightarrow \angle QPR = 50°$



Solution 26: (c) 70°

AB || CD and BC is the transversal.

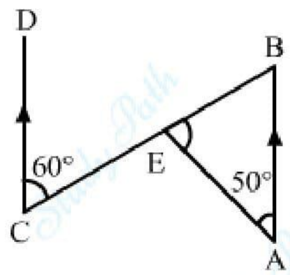
$\therefore \angle ABE = \angle BCD = 60°$ [Alternate Internal Angles]

In ΔABE, we have:

$\angle EAB + \angle ABE + \angle AEB = 180°$ [Sum of the angles of a triangle]

$\Rightarrow 50° + 60° + \angle AEB = 180°$

$\Rightarrow \angle AEB = 70°$



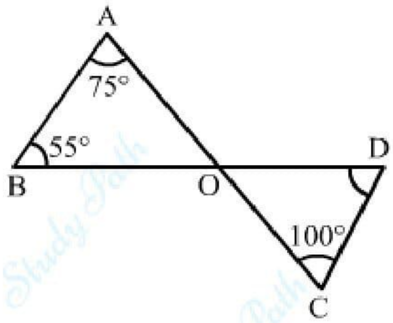
Solution 27: (c) 30°

In ΔOAB, we have:

$\angle OAB + \angle OBA + \angle AOB = 180°$ [Sum of the angles of a triangle]

$\Rightarrow 75° + 55° + \angle AOB = 180°$

$\Rightarrow \angle AOB = 50°$



$\therefore \angle COD = \angle AOB = 50°$ [Vertically-Opposite Angles]

In ΔOCD, we have:

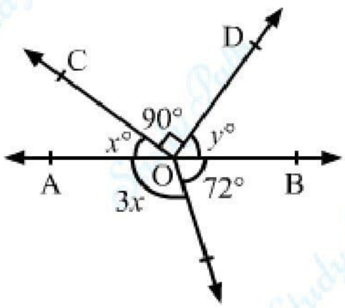
$\angle COD + \angle OCD + \angle ODC = 180°$ [Sum of the angles of a triangle]

$\Rightarrow 50° + 100° + \angle ODC = 180°$

$\Rightarrow \angle ODC = 30°$

Solution 28: (b) 54°

We have:



$3x + 72 = 180°$ [$\because$AOB is a straight line]

$\Rightarrow 3x = 108°$

$\Rightarrow x = 36°$

Also,

$\angle AOC + \angle COD + \angle BOD = 180°$ [$\because$AOB is a straight line]

$\Rightarrow 36° + 90° + y = 180°$

$\Rightarrow 126° + y = 180°$

$\Rightarrow y = 180° - 126°$

$\Rightarrow y = 54°$